

Intergenerational Housing Misallocation and Fertility

Compact Analytical Model with Fertility Sufficient Statistics

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Abstract

Children take up space, and the space they take up is scarce in a particular way: family-sized units are concentrated in the owner-occupied segment of the housing market, while buying owner housing requires a down payment many young households do not have. This note develops a compact stationary two-period OLG theory of that friction. Each period, young households decide whether to enter a housing market, choose between renting and owning, choose how many children to have, and save for old age; births, together with a renewal inflow, generate the next young cohort. Young owners become next period's old incumbents, who decide how much of their carried housing to keep before exiting. The model delivers three results. First, conditional on tenure, prices, and the active constraint, rental size limits and down-payment constraints enter fertility through a common effective family-housing cap. Second, conditional on the current stationary cross-section and fixed dynamic margins, dispersion between constrained young households' and old incumbents' marginal values identifies a local housing-allocation inefficiency; the same capital-gains tax that locks in old incumbents also raises the effective price of owning family-sized space for young interior savers. Third, at fixed prices and a fixed stationary state, a housing policy changes aggregate fertility through constrained exposure, pass-through to the binding cap, the household fertility response, entry, and tenure switching.

The market structure follows [Coven et al. \(2025\)](#): tenure shapes which size limits and financing constraints a household faces, while all floorspace clears in a single market; supply margins are deferred to an extension. Children require housing space, so anything that rations family-sized space raises the private price of children. Family-sized space is rationed by tenure: by the size composition of the rental stock on one side, and by down-payment requirements on the other. The old-side force—capital-gains taxes that fall due on sale but are forgiven at death—amplifies the scarcity and enters young owner decisions through the effective owner price. The basic rental-size-limit and down-payment cap mechanism does not require it.

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Part I

Compact Analytical Model

1 Environment

Agents. The economy is populated by overlapping generations of young and old households. A young household has type

$$i = (y_i, a_i),$$

where $y_i > 0$ is disposable nonhousing income and $a_i \geq 0$ is wealth at entry, inherited from the old who exit as she arrives. She decides whether to enter the housing market and, inside, chooses tenure, occupied housing, fertility, and savings. Old households, indexed by j , come in two groups. Old incumbents are last period's young owners: each enters old age with financial wealth, the house she bought when young, and a possible accrued capital gain; she chooses consumption, how much housing to retain, and how large an estate to leave, then exits. Old renters are last period's young renter entrants; they rent in old age, carry no housing and no gain, and exit. Nonentrants live outside the modeled market in both periods.

Masses. Time is discrete. At the start of period t a mass $M_t > 0$ of potential young households is drawn from a distribution $G_{Y,t}$ over types. Next period's potential young are the maturing children of the current cohort plus a renewal inflow of mass R_t ; entry then determines how many of them participate in the housing market. The period- t measures of old incumbents and old renters are $F_{O,t}$ and $F_{R,t}$. Let $\bar{M}$ denote the stationary cohort mass. In a stationary equilibrium, $G_{Y,t} = G_Y$, $F_{O,t} = F_O$, and $F_{R,t} = F_R$ are reproduced by the transition laws below.

Preferences. Households have preferences over consumption, housing, and children. Children raise the household's consumption and housing needs: each child claims $\chi > 0$ units of the consumption good and $\kappa > 0$ units of housing services, so with consumption spending c_i , occupied housing h_i , and fertility n_i , young preferences are

$$u_i^Y(c_i, h_i, n_i) = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i, \quad \alpha, \vartheta > 0.$$

One more child means less space and less consumption left for the parents: children compete with their parents for the same house and the same budget. Because $\vartheta \log n_i$ rules out $n_i = 0$, every entered household has positive completed fertility: the model studies the intensive margin, not childlessness.

Old preferences are

$$u^O(c_j^O, h_j^O, e_j) = \log c_j^O + \gamma \log h_j^O + \mathcal{B}(e_j), \quad \gamma > 0,$$

over consumption, retained housing, and the estate $e_j \geq 0$ the household leaves when she exits, valued through a warm-glow function $\mathcal{B}$, increasing and concave, possibly identically zero, and common to all old households. The exiting cohort's estates fund the entry wealth of the cohort that arrives, through the transition below. A young household also values her own old age: her objective adds βu^O , evaluated at the old-age choices she will make, with discount factor $\beta \in (0, 1]$; the old state she will hold is generated by the transition below.

Housing. There is a fixed aggregate stock $\bar{H}$ of divisible floorspace, measured in units of housing services. A household occupies floorspace in one of two tenure modes: renting, denoted R , or owning,

denoted O . The modes offer different size limits, the largest sizes available to rent and to own:

$$\mathcal{H}^R = \{h : 0 < h \leq \bar{h}^R\}, \quad \mathcal{H}^O = \{h : 0 < h \leq \bar{h}^O\}, \quad 0 < \bar{h}^R < \bar{h}^O.$$

Family-sized houses are mostly available to buy, not to rent: a household that wants substantially more space than $\bar{h}^R$ must buy it.¹ All occupied floorspace trades in one market. The asset price of a unit of floorspace is P , and a no-arbitrage user-cost condition ties the rental rate to the price:

$$q = \rho P, \quad \rho = r + \delta + \tau^p, \quad (1)$$

where r is the interest rate, δ is depreciation, and τ^p is a recurring property tax. The renter pays q as rent; the owner pays it as mortgage interest, property tax, depreciation, and the forgone return on her equity—different payments, equal per-unit value by arbitrage. Renters and owner-occupiers therefore face the same per-unit flow cost q on new occupancy choices; tenure matters through the size limits and through financing, not through the per-unit price of space. The capital-gains tax below directly shifts old incumbents' retention margin and, when anticipated by interior savers, enters young owners' effective price q^O below. Given q , a higher property tax lowers the asset price P and hence the down payment needed to buy a given amount of space. Throughout, lowercase h denotes an occupancy choice; capital H denotes housing quantities not chosen in the current period—the stock $\bar{H}$, effective caps, carried housing, aggregate demand. At the start of a stationary period, old incumbents carry into old age the houses generated by previous young owner choices, and passive landlords hold the residual stock:

$$H_{\text{pass}} = \bar{H} - \int H_j^0 dF_O(j) \geq 0.$$

In a stationary equilibrium the inequality is implied by market clearing: the previous cohort's owner purchases cannot exceed the stock. Ownership of the stock is thus split between old incumbents and passive landlords: young buyers purchase from incumbents' released units and the landlord stock, renters occupy landlord space, and rents and sale proceeds are transfers. Housing payments are transfers across households, landlords, and fiscal accounts, closed by lump-sum rebates, and reallocating the existing stock costs no real resources. The baseline therefore treats family-sized-unit scarcity as institutional access segmentation within homogeneous floorspace; physical heterogeneity and conversion costs require a multi-stock extension.²

Financing. Buying floorspace requires financing. A young buyer can finance at most a share $\phi \in (0, 1)$ of the purchase; the rest, the down payment $(1 - \phi)Ph$, is cash at closing, posted from wealth in hand: the mortgage is what lenders advance against the house, and they advance nothing against future income, so a_i caps what a buyer can put down. Wealth is otherwise fungible. It enters the young budget below alongside income and finances consumption and saving; buying assets is not expenditure, so the posted equity remains part of the household's wealth. The down-payment inequality is a collateral/access test, not a separate goods expenditure in (7); home equity and

¹Coven et al. (2025) model the same segmentation as a minimum size for owned homes, with rentals available in any size; we cap rental size instead. Both make family-sized space owner-concentrated; ours keeps the renter problem interior at the bottom.

²There is no construction: the model allocates a fixed stock, which is also why the equilibrium is stationary—the demographic laws let population grow, but a fixed stock cannot support constant prices under growth. Two supply extensions are deliberately left out of the baseline and developed later: a price-elastic aggregate supply curve as in Coven et al. (2025), and tenure-specific stocks with separate construction margins, under which the rental and owner user costs diverge endogenously. The local results below survive with q replaced by the relevant tenure-specific user cost.

mortgage debt net out in the old state, leaving $a_{j'} = (1 + r)a'_i - qH_{j'}^0$ and one-period housing services $qH_{j'}^0$. Inheritances therefore move both access to owner housing and lifetime resources. Households save: a young household carries $a'_i \geq 0$ into old age at the exogenous gross return $1 + r$, and cannot borrow unsecured against old age. A renter, with no asset to pledge, therefore cannot borrow at all; a buyer's only debt is the financed share ϕ of the purchase. Entry wealth is assigned from the exiting old's estates by the young-type transition below. The interest rate r is exogenous: households borrow and lend with a passive financial sector, so no domestic capital-market clearing condition is imposed, and the only price determined inside the model is q .

Capital-gains taxation. The model carries two taxes: the property tax τ^p inside the user cost, and the capital-gains tax, whose treatment of death is the old-side institution at the center of the model. Selling a house triggers a tax: the seller pays the rate τ^g on the difference between the sale price and what she paid, in dollars. Dying instead forgives the accrued gain through stepped-up basis; under the estate convention below, the retained physical unit returns tax-free to the stock or passive pool rather than being assigned in kind to a particular entrant. The statutory asset-price tax and its flow-equivalent retention wedge must be kept distinct.

Where do gains come from when the real price never moves? From inflation. Dollar prices and incomes all grow at rate $\varpi > 0$, so relative prices and real choices are unaffected; but the tax form compares dollars from different dates, and a house bought one period ago shows a paper gain even though its real value is unchanged. The tax on that paper gain is real,

$$\ell_P = \tau^g \frac{P \varpi}{1 + \varpi}$$

per unit sold in asset-price units. The old problem is written in one-period goods units, so the object that appears in retention choices is the flow-equivalent wedge

$$\bar{\ell} \equiv \mathbf{a}_\ell \ell_P, \quad \mathbf{a}_\ell > 0,$$

where $\mathbf{a}_\ell$ is the annuitization or period-conversion factor. All formulas below use $\bar{\ell}$, not ℓ_P , and the one-period normalization $\mathbf{a}_\ell = 1$ is only a normalization of units. This is the model's only contact with inflation: budgets, prices, and child requirements are real. Every cohort carries the same gain, so every incumbent's unit bears $\bar{\ell}$ if sold while alive.³ Everything is deterministic: a buyer knows the wedge her future self will face, and the equilibrium-policy section prices that anticipation. An estate tax, by contrast, would fall on the transfer in any form and leave the retention margin undistorted; the gains tax distorts it precisely because death forgives it. Estate-tax instruments live in the quantitative model.⁴

Demography and transitions. The period timing is: potential young types are drawn; young households decide whether to enter; entrants choose tenure, housing, fertility, and savings; old incumbents choose consumption, retained housing, and estates; old households exit; births and the renewal inflow form the next potential-young cohort; current young owners become next period's old incumbents. Let

$$\Gamma_O(dj' \mid i, h, a'; q, \tau^p)$$

be the old-incumbent transition for a current young owner of type i who buys h and carries savings a' . It assigns the house bought when young, $H_{j'}^0 = h$; financial wealth

$$a_{j'} = (1 + r) a' - q H_{j'}^0,$$

³The statutory exclusion, which exempts gains below a dollar threshold, is an institutional detail we suppress; with it, small holdings escape the tax and lock-in sorts by size.

⁴Assessment protection of the Proposition 13 type, under which incumbents are taxed on below-market assessed values until they move, generates the same per-unit retention wedge.

so that total old wealth $a_{j'} + qH_{j'}^0$ is exactly the savings' return—the nominal gain is not real wealth. Housing wealth is in per-period goods units: the old live one more period, so a carried unit is worth one period of services q to its holder, and the shell that outlives her recirculates through the stock. The estate e_j is the nonhousing estate assigned to entrants; retained housing is consumed by the old and then returns tax-free to the stock or passive pool, so qh_j^O is not counted again in Γ_Y . The asset price P appears only in the down-payment requirement. Mortgage balances from youth are netted inside $a_{j'}$: a household that carried a mortgage has lower financial wealth and the same total. The draw does not condition on fertility: estates go to the arriving cohort, not to own children, so the model has no old-age-support motive for births. Young renter entrants map into old-renter states through a parallel kernel $\Gamma_R(dj' \mid i, a')$, which assigns $a_{j'} = (1 + r)a'$; nonentrants receive zero mass under both kernels. Old-renter states follow the same law as (2) with Γ_R and the renter indicator. Given current young choices, the next old-incumbent measure satisfies

$$F_{O,t+1}(A) = M_t \int \pi_{i,t}^E \mathbf{1}\{o_{i,t} = O\} \Gamma_O(A \mid i, h_{i,t}^O, a_{i,t}'^O; q_t, \tau_t^p) dG_{Y,t}(i) \quad (2)$$

for every measurable set A of old states. In stationarity, this measure equals F_O .

Let

$$n_{i,t}^Y = \mathbf{1}\{o_{i,t} = R\} n_{i,t}^R + \mathbf{1}\{o_{i,t} = O\} n_{i,t}^O \quad (3)$$

denote fertility in the chosen inside mode, and let $\bar{n}_i^E \geq 0$ denote fertility in the outside option. The baseline formulas set $\bar{n}_i^E = 0$, but the demographic law allows outside fertility or inflow to renew the population.

Young type $i' = (y_{i'}, a_{i'})$ is generated by fertility, old estates, and the renewal inflow. Let

$$\Gamma_Y(di' \mid i, m, n, F_O, F_R, e; \Upsilon), \quad m \in \{R, O, E\},$$

be the young-type kernel for a birth associated with current type i , mode m , fertility n , old bequest policy e , and renewal law Υ . The old who exit at the end of period t endow the cohort that arrives at $t + 1$: the kernel assigns their estates as the entrants' wealth $a_{i'}$ and supplies the income components of young types. In stationarity, Γ_Y is restricted so that assigned entry wealth aggregates to the estates of all exiting old households, incumbents and renters:

$$\bar{M} \int \left[\pi_i^E n_i^Y \mathbb{E}[a_{i'} \mid i, o_i] + (1 - \pi_i^E) \bar{n}_i^E \mathbb{E}[a_{i'} \mid i, E] \right] dG_Y(i) = \int e_j d(F_O + F_R)(j),$$

where the expectations are over Γ_Y and renewal draws from Υ receive no estates. How estates are assigned across entrants is otherwise unrestricted: equal division, random assignment, and lineage rules are all admissible kernels, and every result below conditions on the realized G_Y ; the quantitative model disciplines the choice by matching the joint distribution of income and inheritances. Let $R_t \geq 0$ be the renewal inflow, outside births or immigration, with type law Υ . The next young cohort satisfies, for every measurable set A of young types,

$$\begin{aligned} M_{t+1} G_{Y,t+1}(A) = M_t \int & \left[\pi_{i,t}^E n_{i,t}^Y \Gamma_Y(A \mid i, o_{i,t}, n_{i,t}^Y, F_{O,t}, F_{R,t}, e_t; \Upsilon) \right. \\ & \left. + (1 - \pi_{i,t}^E) \bar{n}_i^E \Gamma_Y(A \mid i, E, \bar{n}_i^E, F_{O,t}, F_{R,t}, e_t; \Upsilon) \right] dG_{Y,t}(i) + R_t \Upsilon(A). \end{aligned} \quad (4)$$

Setting A equal to the whole young type space gives the cohort-mass law:

$$M_{t+1} = M_t \int [\pi_{i,t}^E n_{i,t}^Y + (1 - \pi_{i,t}^E) \bar{n}_i^E] dG_{Y,t}(i) + R_t. \quad (5)$$

In a stationary equilibrium, $M_t = \bar{M}$, $G_{Y,t} = G_Y$, $F_{O,t} = F_O$, $F_{R,t} = F_R$, and $R_t = \bar{R}$. If $\bar{R} = 0$, stationarity requires average births per potential young household to be one. If $\bar{R} > 0$, it requires average births below one, with the shortfall covered by renewal. Equivalently, with $\bar{M}$ fixed one can set renewal residually, $\bar{R} = \bar{M}(1 - b) \geq 0$, where b is average births per potential young household. With $\bar{R}$ fixed instead, stationarity requires $b < 1$ and $\bar{M} = \bar{R}/(1 - b)$. The propositions below are conditional on a regular stationary equilibrium satisfying one of these closures.

Entry. A potential young household can enter the housing market or take an outside option with lifetime value $\bar{W}^E$; entry is subject to Type-I extreme-value taste shocks with scale $\kappa_E > 0$.

2 The young household's problem

Conditional on entry, household i takes the user cost q and the asset price $P = q/\rho$ as given and chooses a tenure mode $m \in \{R, O\}$, a bundle, and savings. For $m \in \{R, O\}$,

$$W_i^m(q, \tau^p) = \max_{c_i, h_i, n_i, a_i'} \left\{ \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i \right. \\ \left. + \beta [\mathbf{1}\{m = R\} V^R(a_{j'}) + \mathbf{1}\{m = O\} V^O(a_{j'}, h_i)] \right\} \quad (6)$$

subject to

$$c_i + qh_i + a_i' = y_i + a_i, \quad (7)$$

$$h_i \in \mathcal{H}^m, \quad a_i' \geq 0, \quad (8)$$

$$\mathbf{1}\{m = O\} (1 - \phi) P h_i \leq a_i, \quad (9)$$

$$c_i - \chi n_i > 0, \quad h_i - \kappa n_i > 0.$$

V^O and V^R are the old-age values defined in the next section, evaluated at the state the household generates. Nothing here is random: the household knows the old state her choices produce. The current-period market price of space is the same in both modes, wealth is spendable either way, and savings earn $1 + r$ either way. Tenure also matters through the size limit, the owner financing requirement, and the mode-specific old-age continuation state. Children have a market price. Each child claims χ goods and κ units of space, and space rents at q , so providing for a child costs $\chi + \kappa q$ per period; the budget splits into the adults' own consumption and space, the children's cost, and savings. When access to space is constrained, the space part of a child's cost is valued above q —the equilibrium-policy section makes this exact. A useful benchmark is equal scaling, $\chi = \kappa q/\alpha$: the child splits her cost between space and goods in the proportion $\kappa q/\chi = \alpha$, the same proportion in which her parents split their own spending.

Using $P = q/\rho$, owner housing is bounded above by

$$h_i \leq H_i^O(q, \tau^p) = \min \left\{ \bar{h}^O, \frac{a_i \rho}{(1 - \phi)q} \right\}.$$

The renter analog is $H_i^R = \bar{h}^R$. In mode m , H_i^m is the effective family-housing cap. A renter is capped by the rental size limit. A would-be buyer is capped by the owner size limit or by down-payment capacity, whichever is shorter. A policy affects fertility through this channel only when it relaxes the binding component of H_i^m . If the owner mode has no feasible allocation, for example when $a_i = 0$, set $W_i^O = -\infty$.

The inside value of entry is

$$W_i^Y(q, \tau^p) = \max\{W_i^R(q), W_i^O(q, \tau^p)\}. \quad (10)$$

Let $o_i \in \{R, O\}$ denote an optimal tenure.⁵

3 The old household's problem

The baseline young-access mechanism below does not require any old-side force. When the old side is active, the compact model uses the capital-gains institution from the environment. On old-household objects (V^O, V^R, u^O, c_j^O) the superscripts mark the old stage and its tenure; the index j separates old households from young ones. Old incumbent j is last period's young owner and enters with financial wealth a_j (negative when a mortgage balance remains), carried housing H_j^0 , and an accrued gain on it. Old households have no income: resources are returned savings and, for incumbents, the carried house. Each unit she holds faces one choice: sell it now and pay the gains tax, or keep it until death, when the basis steps up and the tax is forgiven. Selling triggers the flow-equivalent tax wedge $\bar{\ell}$ per unit.⁶ She chooses consumption, retained housing, and her estate, with the preferences of Section 1:

$$\begin{aligned} V^O(a_j, H_j^0) &= \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j) \\ \text{subject to } c_j^O + e_j + q h_j^O &= a_j + q H_j^0 - \bar{\ell}(H_j^0 - h_j^O), \quad 0 \leq h_j^O \leq H_j^0. \end{aligned} \quad (11)$$

The old-incumbent state space is restricted to feasible states with positive liquidation wealth,

$$a_j + (q - \bar{\ell})H_j^0 > 0,$$

or equivalently the transition kernel Γ_O puts mass only on such states. Releasing a unit of housing nets $q - \bar{\ell}$, its market value less the tax the sale realizes, while the market values the released space at q . The forgiveness at death is a subsidy to staying put: a tax rule attached to the disposal route, not a preference and not a real resource saving. The planner below counts true old housing utility and true warm-glow bequest utility but does not count the avoided tax as a social benefit. An old household that keeps a large house because she values it is using housing efficiently; an old household that keeps it because selling would realize a taxable gain is not.

An old renter carries no housing and no tax: her wealth is her returned savings, $a_j = (1 + r)a'_i$, positive because zero savings would leave her nothing to consume. With the same preferences she solves

$$V^R(a_j) = \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j) \quad \text{subject to } c_j^O + e_j + q h_j^O = a_j, \quad 0 < h_j^O \leq \bar{h}^R, \quad (12)$$

and at an interior choice $\gamma c_j^O / h_j^O = q$: old rental demand adds to market clearing but carries no wedge.⁷

⁵The hard rental cap $\bar{h}^R$ is the simplest representation of size segmentation. A smooth alternative lets renters pay $R(h) = qh + \varrho(h)$ with $\varrho_h(h) \geq 0$ for large units, so that large rentals exist but at a premium; every marginal condition below goes through with q replaced by $R_h(h)$. Such a premium is also one natural source of a rent-own gap in per-unit costs, which the tenure-switching subsection studies.

⁶The statutory tax is triggered by the sale of the discrete dwelling; with divisible floorspace the per-unit tax is its smooth counterpart after the conversion from ℓ_P to $\bar{\ell}$. Discreteness concentrates the same wedge at the downsizing margin and strengthens retention; it also makes pre-saving plan-contingent, since a strict keeper escapes the bill entirely while a planned downsizer saves the whole of it—the smooth rule is the divisible limit.

⁷Coven et al. (2025) model this lock-in channel in lifecycle form and evaluate a rate reduction. The tax $\bar{\ell}$ prices the

4 Entry

With the entry technology of Section 1, the probability that type i enters the housing market is

$$\pi_i^E(q, \tau^p) = \frac{\exp\{W_i^Y(q, \tau^p)/\kappa_E\}}{\exp\{W_i^Y(q, \tau^p)/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}. \quad (13)$$

The endogenous measure of young entrants is

$$dF_Y^E(i; q, \tau^p) = \bar{M} \pi_i^E(q, \tau^p) dG_Y(i).$$

The limit $\kappa_E \downarrow 0$ gives deterministic entry, with the indifference set $W_i^Y = \bar{W}^E$ assumed to carry zero mass. Conditional on entry, the household chooses tenure according to (10).

5 Competitive equilibrium

A stationary equilibrium clears one aggregate floorspace market and closes the old-incumbent, young-type, and cohort-mass fixed points. Households take the user cost as given, choose tenure and housing, and their choices aggregate into demand for the fixed stock. There is no pairwise assignment of old units to young households: one price equates total demand to the stock, and nothing in that condition asks whether the marginal family-sized unit is held by the household that values it most. That question is deferred to the efficiency analysis below.

Given the policy τ^p , the stock $\bar{H}$, the stationary cohort mass $\bar{M}$, young distribution G_Y , old measures F_O and F_R , and transfer rules, aggregate floorspace demand $H^D(q, \tau^p, x; \bar{M}, G_Y, F_O, F_R)$ at a candidate user cost q adds young households, old incumbents, and old renters:

$$H^D = \bar{M} \int \pi_i^E [\mathbf{1}\{o_i = R\} h_i^R + \mathbf{1}\{o_i = O\} h_i^O] dG_Y(i) + \int h_j^O dF_O(j) + \int h_j^O dF_R(j).$$

In the old-renter integral, h_j^O denotes old-age housing, not owner tenure; it is the solution of (12). Tenure determines which size limit and which financing constraints apply to a household, not which market it buys space in: all of this demand falls on the same stock.

Definition 1 (Stationary competitive equilibrium). *Given $(\tau^p, \bar{H})$, transition laws $(\Gamma_O, \Gamma_R, \Gamma_Y)$, and a stationary renewal law $(\bar{R}, \Upsilon)$, a stationary competitive equilibrium consists of a stationary cohort mass $\bar{M} > 0$, a user cost q , an asset price P , a stationary young distribution G_Y , a stationary old-incumbent measure F_O , a stationary old-renter measure F_R , young mode allocations including savings, tenure choices, entry probabilities, old allocations, and fiscal transfers such that:*

1. conditional on entry, young households solve (6) in each mode, then choose tenure by (10);
2. old incumbents solve (11) and old renters solve (12);
3. entry is given by (13);
4. the asset price satisfies the no-arbitrage condition (1);
5. the single floorspace market clears,

$$H^D(q, \tau^p, x; \bar{M}, G_Y, F_O, F_R) = \bar{H}; \quad (14)$$

same margin statically, and also prices the opposite reform: taxing accrued gains at every transfer, including bequest, removes the step-up and shrinks $\bar{\ell}$ to the value of tax deferral.

6. *passive housing-income, credit, property-tax, capital-gains, and policy-instrument accounts are closed by lump-sum transfers;*
7. *the old-incumbent measure is stationary: for every measurable set A of old states,*

$$F_O(A) = \bar{M} \int \pi_i^E \mathbf{1}\{o_i = O\} \Gamma_O(A \mid i, h_i^O, a_i'^O; q, \tau^p) dG_Y(i), \quad (15)$$

and the old-renter measure satisfies the same condition with Γ_R and $\mathbf{1}\{o_i = R\}$;

8. *the young cohort is stationary: for every measurable set A of young types,*

$$\begin{aligned} \bar{M} G_Y(A) = \bar{M} \int & \left[\pi_i^E n_i^Y \Gamma_Y(A \mid i, o_i, n_i^Y, F_O, F_R, e; \Upsilon) \right. \\ & \left. + (1 - \pi_i^E) \bar{n}_i^E \Gamma_Y(A \mid i, E, \bar{n}_i^E, F_O, F_R, e; \Upsilon) \right] dG_Y(i) + \bar{R} \Upsilon(A), \end{aligned} \quad (16)$$

where e is the old bequest-expenditure policy. Taking A to be the whole young type space gives the stationary cohort-balance condition

$$\bar{M} = \bar{M} \int [\pi_i^E n_i^Y + (1 - \pi_i^E) \bar{n}_i^E] dG_Y(i) + \bar{R}. \quad (17)$$

Let $\Xi^\Pi \geq 0$ denote real resource costs of policy instruments. The transfer rule is chosen so that, after summing entrant and old-household budgets and using (14), the aggregate goods constraint is

$$\begin{aligned} \bar{M} \int & \pi_i^E [\mathbf{1}\{o_i = R\} (c_i^R + a_i'^R - y_i - a_i) + \mathbf{1}\{o_i = O\} (c_i^O + a_i'^O - y_i - a_i)] dG_Y(i) \\ & + \int (c_j^O + e_j - a_j) d(F_O + F_R)(j) + \Xi^\Pi \leq 0. \end{aligned}$$

Housing quantities are disciplined separately by floorspace feasibility: housing payments are transfers that net out against rebates, and the old carried-housing term qH_j^0 is a claim on the existing stock rather than new output. Nonentrants' outside-option consumption and resources are outside this modeled goods account, though their outside fertility can enter the demographic accounting below. The intergenerational books close through the transitions: entry wealth nets against old estates by the adding-up restriction on Γ_Y , and the net return r on carried savings is paid by the passive financial sector, the open-economy counterpart of foreign asset income. Housing in this baseline is a pure allocation problem.

Aggregate births generated by the potential-young cohort are

$$N = \bar{M} \int [\pi_i^E n_i^Y + (1 - \pi_i^E) \bar{n}_i^E] dG_Y(i).$$

In a stationary demographic equilibrium, $N + \bar{R} = \bar{M}$. Under the baseline outside-option normalization $\bar{n}_i^E = 0$, $N = \bar{M} \int \pi_i^E n_i^Y dG_Y(i)$. Everything here is the stationary state: the old cross-section is the one the model itself regenerates, so no separate initial old generation needs specifying; an initial condition would matter only on a transition path, which we do not study.

6 Equilibrium policies

How tightly a constraint binds can be read off the household's bundle: the consumption, housing, and fertility she chooses. A household that is free to adjust equates her marginal value of housing space

to the price of space. A household pinned by a size limit or a financing requirement values space above its price, and the distance between the two is the constraint's bite, measured in consumption goods.

Every condition below is a derivative of one Lagrangian. For mode m ,

$$\begin{aligned}\mathcal{L}_i^m = & \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i + \beta V_{j'} \\ & + \lambda_i^m (y_i + a_i - c_i - q h_i - a'_i) + \theta_i^m (\bar{h}^m - h_i) + \xi_i a'_i \\ & + \mathbf{1}\{m = O\} \mu_i (a_i - (1 - \phi) P h_i),\end{aligned}$$

with λ_i^m on the budget (7), $\theta_i^m \geq 0$ on the size limit, and $\mu_i \geq 0$ on the down-payment constraint (9), active only for buyers. The generated old state (8) is substituted into $V_{j'}$, and $\xi_i \geq 0$ is the multiplier on the no-borrowing bound $a'_i \geq 0$ —for a renter, the constraint that she cannot borrow at all; a buyer's only debt is the mortgage. The domain conditions hold strictly at any optimum.

For a renter ($m = R$), at an interior allocation with respect to consumption and fertility,

$$\begin{aligned}\frac{1}{c_i^R - \chi n_i^R} &= \lambda_i^R, \\ \frac{\alpha}{h_i^R - \kappa n_i^R} &= \lambda_i^R q + \theta_i^R, \\ \frac{\vartheta}{n_i^R} &= \frac{\chi}{c_i^R - \chi n_i^R} + \frac{\alpha \kappa}{h_i^R - \kappa n_i^R}.\end{aligned}\tag{18}$$

Converting the cap multiplier into consumption goods, $\zeta_i^R = \theta_i^R / \lambda_i^R \geq 0$, the renter's marginal value of housing is

$$\frac{u_{h,i}^Y}{u_{c,i}^Y} = \frac{\alpha (c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = q + \zeta_i^R.\tag{19}$$

A renter against the size cap values one more unit of space above the rent it would command, and the gap is computable from the observed bundle as $\zeta_i^R = \alpha (c_i^R - \chi n_i^R) / (h_i^R - \kappa n_i^R) - q$, the household's tradeoff between space and consumption, net of children, in excess of the market rent. An unconstrained renter has $\zeta_i^R = 0$.

Assumption 1 (Regularity at the old and saving margins). *The flow-equivalent gains wedge satisfies $0 \leq \bar{\ell} < q$. The generated old owner state has positive liquidation wealth, $a_{j'} + (q - \bar{\ell}) H_{j'}^0 > 0$, and an interior retention choice, $0 < h_{j'}^O < H_{j'}^0$. The young household is an interior saver, $a'_i > 0$. Where old-renter first-order conditions or the tenure-switching constant are used, the corresponding old-stage choices are interior.*

Anticipation is a price here, not a friction to assume away. Buying one more unit raises anticipated old wealth by its housing value and costs that value today; it also commits the household to the tax $\bar{\ell}$ on the marginal unit her old self sells. By the envelope condition $\partial V^O / \partial H^0 = (q - \bar{\ell}) / c^O$, which the assumption keeps smooth, and by the generated owner state $a_{j'} = (1 + r) a'_i - q h_i$ and $H_{j'}^0 = h_i$,

$$\frac{\partial V^O}{\partial h_i} = -q V_a^O + V_H^O = -\frac{q}{c_{j'}^O} + \frac{q - \bar{\ell}}{c_{j'}^O} = -\frac{\bar{\ell}}{c_{j'}^O}.$$

The continuation therefore contributes $-\beta \bar{\ell} / c_{j'}^O$ to the buyer's housing condition, and the saving condition converts it into goods:

$$\frac{\beta \bar{\ell} (c_i - \chi n_i)}{c_{j'}^O} = \frac{\bar{\ell}}{1 + r}.$$

A buyer therefore pays an effective price

$$q^O \equiv q + \frac{\bar{\ell}}{1+r},$$

per unit of space—the market price plus her own discounted lock-in tax, pre-funded through savings rather than paid as a flow—while a renter pays q . The tax acts twice: ex post it locks in the current old, and ex ante it taxes ownership of family-sized space. For a household against the unsecured borrowing bound, the equality above is replaced by the type-specific continuation price $q_i^O = q + \beta\bar{\ell}/(\lambda_i c_{j'}^O)$; the closed-form q^O and the measured owner financing gap below use Assumption 1.

For a young owner ($m = O$), at an interior allocation with respect to consumption and fertility,

$$\begin{aligned} \frac{1}{c_i^O - \chi n_i^O} &= \lambda_i^O, \\ \frac{\alpha}{h_i^O - \kappa n_i^O} &= \lambda_i^O q + \frac{\beta \bar{\ell}}{c_{j'}^O} + \mu_i(1 - \phi)P + \theta_i^O, \\ \frac{\vartheta}{n_i^O} &= \frac{\chi}{c_i^O - \chi n_i^O} + \frac{\alpha \kappa}{h_i^O - \kappa n_i^O}. \end{aligned} \tag{20}$$

The same conversion for the owner collects the access constraints, each priced by the primitive that generates it:

$$\zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i^O}(1 - \phi)P}_{\zeta_i^{O,F}, \text{ down-payment component}} + \underbrace{\frac{\theta_i^O}{\lambda_i^O}}_{\text{owner-size-limit component}} \geq 0.$$

The first term is the collateral margin: it is positive when the household's wealth a_i cannot cover the down payment $(1 - \phi)P$ on a larger unit, and F stands for financing. The second is the owner size limit. A payment-to-income test would add a parallel financing component to $\zeta_i^{O,F}$; the quantitative model carries one. If the owner size cap does not bind, then $\theta_i^O = 0$ and $\zeta_i^O = \zeta_i^{O,F}$: over and above her effective price, the household is constrained by finance, not by the stock. The young owner's marginal value of owner housing is

$$\frac{u_{h,i}^Y}{u_{c,i}^Y} = \frac{\alpha(c_i^O - \chi n_i^O)}{h_i^O - \kappa n_i^O} = q^O + \zeta_i^O. \tag{21}$$

Both tenure modes share the saving margin

$$\frac{1}{c_i - \chi n_i} = \beta(1+r) \frac{1}{c_{j'}^O} + \xi_i, \quad \xi_i \geq 0, \quad \xi_i a_i' = 0, \tag{22}$$

the first-order condition for saving, with $c_{j'}^O$ old consumption at the generated state: both old problems have envelope $\partial V/\partial a_j = 1/c_{j'}^O$, carried savings pay $1+r$, and $\xi_i > 0$ exactly when the household would like to borrow against old age and cannot. In the benchmark it pins savings in closed form. For $b > 0$, let $\mathcal{B}(e) = b \log e$ and take the bequest margin interior; for $b = 0$, take $\mathcal{B} \equiv 0$ and allow $e = 0$. With old choices interior, old consumption is proportional to liquidation wealth, and the condition collapses to the savings rule

$$a_i' = \beta(1 + \gamma + b)(c_i - \chi n_i) + \mathbf{1}\{m = O\} \frac{\bar{\ell} h_i}{1+r}, \tag{23}$$

with γ the old housing weight and b the warm-glow weight; renters drop the second term. In the first term the gross return drops out, the usual log cancellation; the second is the discounted future tax bill, which the household saves in advance: anticipating the lock-in tax is itself a savings motive. In the interior divisible formulation, the tax is equivalent to a certain wealth cost $\bar{\ell}h_i$ at the generated old state; statutory payments still occur only on sold units. At an interior retention choice a kept unit costs its forgone net-of-tax sale value, so the tax lowers the old budget by $\bar{\ell}h_i$ whatever the keep-sell split; keeping changes what the budget is spent on, not its size. Only a household that would keep the whole house no matter what escapes the bill; Assumption 1's interior retention rules her out. The savings load $k \equiv \beta(1 + \gamma + b)$ is old age's claim on each unit of adult consumption; the fertility comparative statics use it as shorthand.

For an old owner at an interior choice of current housing to keep,

$$\frac{u_{h,j}^O}{u_{c,j}^O} = \frac{\gamma c_j^O}{h_j^O} = q - \bar{\ell}. \quad (24)$$

The old owner's valuation sits *below* the market price by exactly the tax: she keeps marginal space she values at less than the market would pay, because selling it would realize the taxable gain. If bequests are interior, then

$$\mathcal{B}'(e_j) = \frac{1}{c_j^O},$$

which matches the planner's bequest condition: leaving wealth to children is a preference the model respects, not a distortion. An old renter at an interior choice has $u_{h,j}^O/u_{c,j}^O = q$ exactly; her demand is wedge-free.

Proposition 1 (Constrained family-housing access and fertility). *Fix an entered young household and its optimal tenure mode $m \in \{R, O\}$, with $q^R = q$ and, under Assumption 1, $q^O = q + \bar{\ell}/(1 + r)$. The solved policy gives the level of fertility; the first-order condition behind that policy shows where housing enters. In goods units,*

$$\underbrace{\frac{\vartheta(c_i^m - \chi n_i^m)}{n_i^m}}_{\text{willingness to pay for a marginal child}} = \underbrace{\chi + \kappa(q^m + \zeta_i^m)}_{\text{full price of a child}}.$$

Thus rental size limits and owner access constraints enter fertility through the housing services required per child.

Proof. For renters, multiply (18) by $1/\lambda_i^R = c_i^R - \chi n_i^R$ and use (19). For owners, multiply (20) by $1/\lambda_i^O = c_i^O - \chi n_i^O$ and use (21). $\square$

A child's space requirement is priced at the household's *own* shadow value of space, not at the market price. In an unconstrained mode, the full price of a child is $\chi + \kappa q^m$. For a constrained household it is $\chi + \kappa(q^m + \zeta_i^m)$, so a binding access constraint acts as a tax of $\kappa \zeta_i^m$ per child. Children become more expensive exactly where family-sized space is hard to reach, even with market prices unchanged. Two limits anchor the framework: as $H_i^m \rightarrow \infty$ every household pays the unconstrained child price $\chi + \kappa q^m$, and with $\kappa = 0$ children claim no space and the price reduces to the standard child-goods cost χ . On the financing side, when the collateral component binds, $\partial H_i^O / \partial \tau^p = a_i / ((1 - \phi)q) > 0$: a higher property tax lowers the asset price at a given user cost and relaxes the down payment, the capitalization channel of Coven et al. (2025). This

derivative isolates the down-payment capitalization channel holding $\bar{\ell}$ fixed; if the statutory liability $\ell_P = \tau^g P \varpi / (1 + \varpi)$ and the conversion factor $\mathbf{a}_\ell$ make $\bar{\ell} = \mathbf{a}_\ell \ell_P$ move with P , the policy also changes q^O and the old retention wedge.

If no access or size limit constraint binds in the relevant tenure mode, the corresponding access value is zero. For example, an unconstrained young owner satisfies

$$\begin{aligned} c_i^u - \chi n_i^u &= \frac{y_i + a_i}{1 + \alpha + \vartheta + k}, & a_i^u &= k(c_i^u - \chi n_i^u) + \mathbf{1}\{m = O\} \frac{\bar{\ell} h_i^u}{1 + r}, \\ n_i^u &= \frac{\vartheta(y_i + a_i)}{(1 + \alpha + \vartheta + k)(\chi + \kappa q^O)}, \\ h_i^u &= \frac{\alpha(y_i + a_i)}{(1 + \alpha + \vartheta + k)q^O} + \kappa \frac{\vartheta(y_i + a_i)}{(1 + \alpha + \vartheta + k)(\chi + \kappa q^O)}. \end{aligned}$$

The renter analog has the same closed form and is feasible only if $h_i^u \leq \bar{h}^R$.

7 Same-mass planner benchmark

The centralized comparison is evaluated against a stationary competitive equilibrium. The planner has the same preferences, fixed housing stock, tenure size limits, stationary young distribution, old measures, and transition laws as the competitive economy. It is fertility-neutral: births enter because parents value children and children use goods and space, not because the planner has a separate social target for the number of births.

The mass convention matters. Across stationary same-mass comparisons, the renewal/outside inflow is the residual object that keeps the potential-young mass $\bar{M}$ and type law G_Y fixed. This removes the mechanical utilitarian gain from creating more households and isolates the housing-allocation margin.

For the local variations below, the stationary cohort mass $\bar{M}$, young distribution G_Y , old measures F_O and F_R , transition laws $(\Gamma_O, \Gamma_R, \Gamma_Y)$, renewal inflow $\bar{R}$, and tax rules are held fixed at the instant of the variation because the demographic fixed point is not being re-solved. The question is which constraints young families face as real scarcity and which are financing arrangements that a different institutional design could undo.

The planner chooses inside allocations, tenure assignments, young consumption, housing and fertility, old consumption, housing and estates, and policy instruments for the current stationary cross-section. It does not move the savings margin: young savings a_i^u , and the old wealth they generate, are held at their equilibrium values, so the comparison is within-period reallocation given the intergenerational wealth transition. The planner respects floorspace feasibility and the rental and owner size limits.

The planner faces the economy's real constraints and nothing else: the floorspace stock, the tenure size limits, goods feasibility, and the domain conditions below. It does not face household budgets, since allocations are supported by lump-sum transfers, and it does not face the financing inequalities, which restrict market purchases rather than occupancy. The planner counts true old housing utility and true warm-glow bequest utility; it does not count the avoided gains tax $\bar{\ell}$, which is a transfer.

Let $\omega_i^Y > 0$ and $\omega_j^O > 0$ be Pareto weights. The main benchmark holds the competitive entry probabilities π_i^E fixed and reallocates the current stationary cross-section. For types with $\pi_i^E = 0$, the planner's inside allocation is irrelevant. For types with $\pi_i^E > 0$, the planner may assign a tenure $\alpha_i^P \in \{R, O\}$ and an allocation (c_i, h_i, n_i) .

The planner solves

$$\max \bar{M} \int \omega_i^Y \pi_i^E u_i^Y(c_i, h_i, n_i) dG_Y(i) + \int \omega_j^O u_j^O(c_j^O, h_j^O, e_j) d(F_O + F_R)(j) \quad (25)$$

subject to floorspace feasibility,

$$\bar{M} \int \pi_i^E h_i dG_Y(i) + \int h_j^O d(F_O + F_R)(j) \leq \bar{H},$$

size-limit feasibility,

$$h_i \in \mathcal{H}^R \text{ if } o_i^P = R, \quad h_i \in \mathcal{H}^O \text{ if } o_i^P = O,$$

and goods feasibility,

$$\begin{aligned} \bar{M} \int \pi_i^E (c_i + a'_i - y_i - a_i) dG_Y(i) \\ + \int (c_j^O + e_j - a_j) d(F_O + F_R)(j) + \Xi^\Pi \leq 0, \end{aligned}$$

All allocations also satisfy

$$c_i - \chi n_i > 0, \quad n_i > 0, \quad h_i - \kappa n_i > 0,$$

for young households, and

$$c_j^O > 0, \quad e_j \geq 0, \quad 0 \leq h_j^O \leq H_j^0,$$

for old incumbents, with the retention bound replaced by the rental size limit for old renters.

Let Λ be the multiplier on goods feasibility and ΛQ the multiplier on floorspace feasibility, so that Q is the planner's scarcity price of space in consumption goods. For a fixed tenure assignment and active set, and away from size-limit corners, an interior planner allocation satisfies

$$\begin{aligned} \frac{u_{h,i}^Y}{u_{c,i}^Y} &= Q && \text{for young households,} \\ \frac{u_{h,j}^O}{u_{c,j}^O} &= Q && \text{for old housing,} \\ \frac{\vartheta(c_i - \chi n_i)}{n_i} &= \chi + \kappa Q, \\ \frac{\mathcal{B}'(e_j)}{u_{c,j}^O} &= 1 && \text{if } e_j > 0. \end{aligned}$$

The planner equalizes every household's housing-consumption tradeoff to the common scarcity price of space, prices children at $\chi + \kappa Q$, and respects the bequest motive. The fixed entry weight π_i^E multiplies both the young objective and the young resource use, so it cancels from the housing-consumption and fertility ratios for every type with positive entry probability.

Conditional on the entry measure and tenure assignment, the planner's interior allocation solves in one line. With no direct social birth payoff, and with $\mathcal{B}(e) = b \log e$ for $b > 0$ and an interior bequest margin, or $\mathcal{B} \equiv 0$ for $b = 0$,

$$c_i - \chi n_i = \frac{\omega_i^Y}{\Lambda}, \quad h_i - \kappa n_i = \frac{\alpha \omega_i^Y}{\Lambda Q}, \quad n_i = \frac{\vartheta \omega_i^Y}{\Lambda(\chi + \kappa Q)},$$

$$c_j^O = \frac{\omega_j^O}{\Lambda}, \quad h_j^O = \frac{\gamma \omega_j^O}{\Lambda Q}, \quad e_j = \frac{b \omega_j^O}{\Lambda},$$

with (Λ, Q) pinned by goods and floorspace feasibility. The weights are not innocent: ω scales each household's consumption and space together, so any statement about who gets more space under the planner than in equilibrium depends on the weights chosen. What no weight can touch is the ratio: every young planner household satisfies

$$\frac{h_i - \kappa n_i}{c_i - \chi n_i} = \frac{\alpha}{Q},$$

common across young households under any ω . Old planner households separately satisfy $h_j^O/c_j^O = \gamma/Q$. The equilibrium disperses the young ratio to $\alpha/(q^m + \zeta_i^m)$ across constrained young households and, for old incumbents, $\gamma/(q - \bar{\ell})$ against the planner's γ/Q . Dispersed ratios cannot be matched by any choice of weights. The local efficiency test below states this violation constructively: between a high-ratio and a low-ratio household, goods are on the table.

Remark 1 (Fully endogenous population composition). *If the planner were allowed to choose the stationary population measure itself, marginal fertility would also change the future entrant pool and the assignment of inherited wealth. Additional transition shadow values would then enter the fertility first-order condition. That is a different planner problem. The same-mass benchmark here isolates the housing-allocation mechanism and leaves those demographic costates out of the main text.*

8 Efficiency

At a stationary competitive equilibrium, the price makes aggregate demand equal the fixed stock. Market clearing alone is not enough for efficiency. Conditional on the stationary cross-section and transition margins, efficiency also requires that no feasible reassignment would move marginal space toward households who value it more. The test below is compensated: reallocate a unit, hold every household at its prior utility with lump-sum transfers, and ask whether goods are left over. A positive residual is a Pareto improvement under any welfare weights for this fixed-margin compensated variation—it restates the planner ratio violation constructively, harvesting the dispersion that no weight choice can reconcile.

Proposition 2 (Planner-equilibrium comparison, local). *Consider an interior stationary competitive equilibrium and a fertility-neutral planner with the same preferences, housing stock, tenure size limits, stationary young distribution, old measures, and transition laws held fixed at the instant of the variation. Suppose policy accounts are resource-neutral. Conditional on the equilibrium entry measure and tenure assignments, and away from real size-limit corners, the competitive allocation solves the planner's conditional allocation problem only if no feasible marginal reallocation of existing floorspace has positive goods-equivalent surplus. The variation below resizes two owner-occupied houses and converts no tenures.*

In the zero-real-cost direct reassignment case, for a young owner whose owner size limit is slack and whose financial constraints bind, and for an old incumbent at an interior retained-housing choice, the planner's marginal surplus from increasing the young owner's space by one unit and reducing

the old incumbent's retained space by one unit is

$$\underbrace{(q^O + \zeta_i^{O,F})}_{\text{buyer's marginal value}} - \underbrace{(q - \bar{\ell})}_{\text{incumbent's marginal value}} = \zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell} > 0 \quad \text{for an improving reallocation.}$$

The buyer's anticipated tax, like the incumbent's realized one, is not a real resource cost and can offset transfers in the compensated allocation. Thus any such pair with $\zeta_i^{O,F} + \bar{\ell}/(1+r) + \bar{\ell} > 0$ prevents the competitive allocation from solving the planner's conditional allocation problem. A positive young financial access gap and a positive retention tax $\bar{\ell}$ mark an inefficiency whenever the released space has a feasible recipient whose marginal value exceeds the old incumbent's true marginal value.

Conversely, if the financial implementation gaps are zero for all relevant young owners and $\bar{\ell} = 0$ for old incumbents at interior retention choices, then, conditional on entry, tenure assignments, savings, and real size limit constraints, the competitive allocation satisfies the planner's conditional housing and fertility first-order conditions for suitable Pareto weights.

If the same reallocation requires marginal real implementation cost r_i^F , that cost is subtracted from the surplus; if the constraint stands for an enforcement primitive no instrument relaxes, the variation is infeasible. The display reads term by term. The buyer's marginal value is her effective price plus the financing gap. The incumbent's marginal value is the net-of-tax sale price: what releasing one more unit gives her. The market price q cancels. What remains are the financing gap, the buyer's anticipated capital-gains tax, and the incumbent's realized capital-gains tax. The two tax terms are not resource costs and can offset transfers in the compensated allocation.

Proof. The result is a direct marginal-variation comparison that preserves aggregate floorspace feasibility. Increase household a 's occupied floorspace by dh and reduce household b 's occupied floorspace by the same dh , holding entry, tenure assignments, active real size limits, fertility, savings, and stationary transition margins fixed at the instant of the variation. In the direct-reassignment benchmark, the goods-equivalent first-order welfare effect is

$$\frac{u_{h,a}}{u_{c,a}} - \frac{u_{h,b}}{u_{c,b}}.$$

For a young owner with a slack owner size limit and binding financial implementation constraints,

$$\frac{u_{h,i}^Y}{u_{c,i}^Y} = q^O + \zeta_i^{O,F}.$$

The anticipated-tax component $\bar{\ell}/(1+r)$ of q^O is not a real resource cost. For an old incumbent at an interior retention choice,

$$\frac{u_{h,j}^O}{u_{c,j}^O} = q - \bar{\ell}.$$

Subtracting gives

$$\zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell}.$$

A positive value is a feasible first-order improvement: transfer goods to hold each household at its pre-variation utility, and the residual $\left[\zeta_i^{O,F} + \bar{\ell}/(1+r) + \bar{\ell}\right] dh$ is available as surplus. Therefore the competitive allocation cannot solve the planner's conditional allocation problem. If relaxing the young household's financial implementation constraints has marginal real resource cost r_i^F , the residual is reduced by $r_i^F dh$.

If all implementation-relevant financial gaps and retention taxes are zero, the competitive marginal values of all households away from real size-limit corners coincide with the common market user cost q . Real size-limit corners are respected by both the competitive economy and the constrained planner. With no direct social birth payoff, the competitive fertility condition then matches the planner's fertility condition evaluated at the same scarcity price, and Pareto weights can be chosen to support the competitive consumption allocation. $\square$

The proposition does not characterize the efficiency of the entry, tenure-assignment, or intergenerational transition margins themselves; those margins are held fixed. It also does not say that every cap is a market failure: real size limits remain real constraints, while financial gaps matter only to the extent that an implementable instrument can reach them at a low enough real resource cost.

The misallocation has two sides. A young household that cannot post a larger down payment values the marginal unit of space at $q^O + \zeta_i^{O,F}$, above the market value of that space; the market clears, yet she cannot buy more of it. An old incumbent sitting on a taxable gain keeps the marginal unit while valuing it at $q - \bar{\ell}$, below the market value of that space. Moving one unit from the second household to the first would raise welfare by $\zeta_i^{O,F} + \bar{\ell}/(1+r) + \bar{\ell}$, measured in consumption goods, before any real implementation cost. Both pieces are measurable in principle given the preference parameters and the maintained margin assumptions: the young gap from bundles, via $\zeta_i^{O,F} = \alpha(c_i^O - \chi n_i^O)/(h_i^O - \kappa n_i^O) - q - \bar{\ell}/(1+r)$ when the size cap is slack and the saver/old-retention margins are interior—the bundle gap over market rent net of the known tax term—and the old gap from deeds and the tax schedule, via the purchase basis, current value, statutory exclusions, marginal tax rate, and sale-versus-bequest treatment. The old-side force amplifies the misallocation but is not needed for it: the young-access gap is positive whenever a purchase constraint binds with positive shadow value, with or without the gains tax.

Two qualifications matter. First, old occupancy is not inefficient by itself. An old household that keeps a large house because she values it, or because she intends to leave it, is using housing efficiently; those motives are inside u^O and the planner counts them. The inefficiency is only the part of retention supported by the forgiveness at death. Second, not every binding constraint is an inefficiency. The size limits are real features of the housing stock: when a renter is pinned at $\bar{h}^R$, the planner—facing the same stock—is pinned with her, and the multiplier on the size limit is the shadow value of a real constraint, not a market failure. Changing the size limits requires construction, conversion, or reassignment instruments, whose resource costs belong in Ξ^{Π} . Down-payment requirements are financing arrangements; their efficiency cost depends on whether feasible instruments can reach them.

A worked example. A two-type local allocation puts numbers on every object above. Normalize $q = 1$ and set $\rho = 0.25$, so $P = 4$. The young household has $y = 1$, $a = 0.20$, $\alpha = 0.5$, $\vartheta = 0.4$, $\kappa = 0.3$, and the equal-scaling child cost $\chi = \kappa q/\alpha = 0.6$; financing is $\phi = 0.80$; the size limits are $\bar{h}^R = 0.15$ and $\bar{h}^O = 1$. The flow-equivalent gains wedge is $\bar{\ell} = 0.2$ per unit; the period return is $r = 0.35$, so the anticipated-tax wedge is $\bar{\ell}/(1+r) = 0.148$ and the buyer's effective price is $q^O = 1.148$; the discount factor is $\beta = 0.1763$, so with $\gamma = 0.5$ and no bequest motive the savings load is $k = \beta(1+\gamma) = 0.264$. The old incumbent has $a_j = 0.2$ and carried housing $H^0 = 0.25$. The

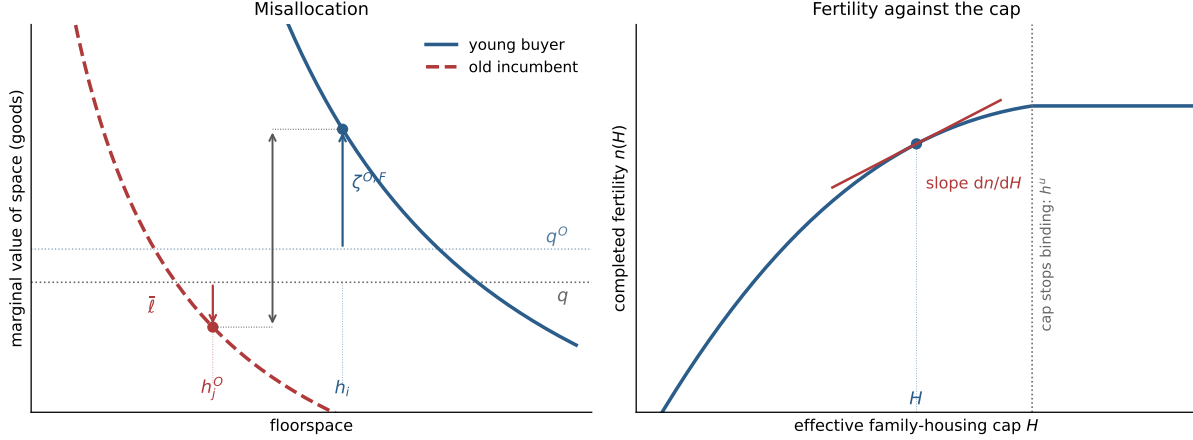


Figure 1: The worked local allocation. Left: marginal value of space. The collateral-capped young owner sits at $h = 0.25$ valuing space at $q^O + \zeta^{O,F} = 1.68$; the incumbent retains $h^O = 1/6$ valuing it at $q - \bar{\ell} = 0.80$; moving a marginal unit from her to the buyer frees 0.88 in goods. Right: completed fertility against the effective cap. At the cap the slope is $dn/dH = 0.30$, and the cap stops binding at $h^u = 0.31$.

stock $\bar{H} = 5/12$ closes the illustrated cross-section at $q = 1$; entry is deterministic with the inside value above the outside option.

The young household buys. At the purchase/flow level both modes face q ; after the anticipated owner tax, owner space has effective price q^O , and at these parameters the larger owner cap still makes ownership optimal. The owner cap is $H^O = \min\{1, a\rho/((1-\phi)q)\} = \min\{1, 0.25\} = 0.25$, the collateral component, against the rental size limit 0.15. The cap binds—the unconstrained optimum is $h^u = 0.312$ —so $h = 0.25$. The budget is $c + qh + a' = y + a = 1.20$; the savings rule gives $a' = k(c - \chi n) + \bar{\ell}h/(1+r) = 0.163 + 0.037 = 0.20$, so the buyer carries exactly her initial wealth, and the fertility condition gives $n^* = 0.223$, with adult consumption $c - \chi n = 0.616$ and adult space $h - \kappa n = 0.183$. The bundle reveals a gap of $\alpha(c - \chi n)/(h - \kappa n) - q = 0.68$ over market rent: 0.148 of it is her own discounted future tax, and the rest, $\zeta^{O,F} = 0.53$, is financing—she would pay above her effective price for one more unit of space and cannot post the down payment to get it. The incumbent's released units net $q - \bar{\ell}$, so her liquidation wealth is $a_j + (q - \bar{\ell})H^0 = 0.40$ and she keeps $h^O = 1/6$ at the interior solution of $\gamma c_j^O/h_j^O = q - \bar{\ell}$, valuing retained space at 0.80; her released $1/12$ plus the passive stock $1/6$ house the young buyer, and the cross-section clears at $\bar{H} = 5/12$. The incumbent state is taken as given for this local allocation; the buyer anticipates her own future $\bar{\ell}$ through q^O . Figure 1 draws the two sides.

Every object can be read off this one allocation. The left panel is a compensated trade: pay the incumbent for a marginal unit at her marginal value, $q - \bar{\ell} = 0.80$; charge the buyer at hers, $q^O + \zeta^{O,F} = 1.68$; both households are exactly as well off, and the residual $\zeta^{O,F} + \bar{\ell}/(1+r) + \bar{\ell} = 0.88$ is left in goods. The cap derivative is $dn/dH = 0.30$; and a down-payment grant passes through at $\partial H^O/\partial a_i = \rho/((1-\phi)q) = 1.25$, so a wealth transfer of 0.01 raises the cap by 0.0125 and completed fertility by 0.0038 through access, plus 0.0012 through the budget now that wealth is spendable, 0.0050 in all. The gap, the tax, and the responses are read off the bundle, the tax schedule, and the financing terms.

9 Fertility and entry

The fertility implication does not require a planner who directly values births. It follows from the private household problem. Children require consumption and housing services, so a constraint on family-capable housing can change fertility even when fertility enters only parental utility.

Fix a tenure mode and suppress the type index. Take a household whose housing is pinned at the effective family-housing cap defined in Section 2: the rental size limit for a renter, and the shorter of the owner size limit and the collateral bound for a buyer. Write H for the level of that cap. The experiment is to relax the cap by one unit of space and study completed fertility, differentiating with respect to the cap level while holding the other mode primitives fixed. For renters, $H = \bar{h}^R$. For young owners,

$$H = \min \left\{ \bar{h}^O, \frac{a_i \rho}{(1 - \phi)q} \right\}.$$

Holding $(w, \chi, q^m, \alpha, \vartheta, \kappa, k)$ fixed, where $w \equiv y + a$ is lifetime resources and q^m is the mode's effective price of space— q for renters, and $q + \bar{\ell}/(1 + r)$ for interior-saver buyers under Assumption 1—the mode problem adds savings to the bundle choice. Concentrating the savings rule into the objective and the budget reduces it to

$$\max_{c, h, n} \{ (1 + k) \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta \log n \}$$

up to a type constant, subject to

$$(1 + k)(c - \chi n) + \chi n + q^m h = w, \quad h \leq H, \quad c - \chi n > 0, \quad h - \kappa n > 0 :$$

the static problem with consumption weight $1 + k$, resources w , and the mode price q^m . Each unit of adult consumption commits $1 + k$ units of resources, one spent now and k carried for old age; each unit of owner space commits its market cost now plus the discounted tax, saved in advance. The unconstrained mode allocation is

$$\begin{aligned} c^u - \chi n^u &= \frac{w}{1 + \alpha + \vartheta + k}, & n^u &= \frac{\vartheta w}{(1 + \alpha + \vartheta + k)(\chi + \kappa q^m)}, \\ h^u &= \frac{\alpha w}{(1 + \alpha + \vartheta + k) q^m} + \kappa \frac{\vartheta w}{(1 + \alpha + \vartheta + k)(\chi + \kappa q^m)}. \end{aligned}$$

Suppose $0 < H < h^u$, so the housing cap binds. Then $h(H) = H$. Write adult residual consumption and adult residual space as

$$x \equiv c - \chi n > 0, \quad s \equiv H - \kappa n > 0.$$

These are the allocation objects that enter both the child-price equation and the cap-response formula. Since the resource residual in the reduced problem is

$$A \equiv w - q^m H - \chi n = (1 + k)x,$$

the constrained fertility choice $n(H)$ is the unique solution to

$$\frac{\vartheta}{n} = \frac{(1 + k) \chi}{w - q^m H - \chi n} + \frac{\alpha \kappa}{H - \kappa n}.$$

Implicit differentiation gives

$$\frac{dn}{dH} = \frac{\frac{\alpha\kappa}{(H-\kappa n)^2} - \frac{(1+k)\chi q^m}{(w-q^m H - \chi n)^2}}{\frac{\vartheta}{n^2} + \frac{(1+k)\chi^2}{(w-q^m H - \chi n)^2} + \frac{\alpha\kappa^2}{(H-\kappa n)^2}}.$$

Equivalently, in residual bundle objects, define

$$\mathcal{D}^m \equiv \frac{\vartheta}{n^2} + \frac{\chi^2}{(1+k)x^2} + \frac{\alpha\kappa^2}{s^2} > 0.$$

Then the intensive fertility response to one more unit of permitted family housing is

$$\Psi^m \equiv \frac{\partial n}{\partial H} = \frac{\frac{\alpha\kappa}{s^2} - \frac{\chi q^m}{(1+k)x^2}}{\mathcal{D}^m}.$$

The numerator has two parts. Extra space relaxes the child's housing requirement, raising fertility through $\alpha\kappa/s^2$; paying for the larger unit tightens the budget around the child's goods requirement, lowering fertility through $\chi q^m/[(1+k)x^2]$. The denominator is positive curvature, so the sign is determined by the numerator.

The sharp household-specific sign condition is

$$\Psi^m > 0 \iff \chi < \frac{(1+k)\alpha\kappa x^2}{q^m s^2}.$$

Using the bundle-implied housing-access gap,

$$\zeta^m = \frac{\alpha x}{s} - q^m,$$

the same condition can be written as

$$\Psi^m > 0 \iff \chi < \frac{(1+k)\kappa(q^m + \zeta^m)^2}{\alpha q^m}.$$

Thus a more tightly space-constrained household is more likely to raise fertility when the cap relaxes.

This condition has a direct Stone–Geary interpretation. At an unconstrained interior mode allocation,

$$\frac{q^m(h^u - \kappa n^u)}{c^u - \chi n^u} = \alpha,$$

unchanged by the savings margin and evaluated at the mode price. The child input bundle has the same housing-to-consumption expenditure ratio as the adults' own bundle when

$$\frac{q^m \kappa}{\chi} = \alpha, \quad \text{equivalently} \quad \chi = \frac{\kappa q^m}{\alpha},$$

the equal-scaling benchmark at mode prices. The earlier market-price benchmark, $\chi = \kappa q/\alpha$, is the renter or no-tax version; for owners with $q^m = q^O$ it is distinct from lifetime equal scaling. With savings, a simple uniform sufficient condition for a positive response at every strictly binding cap is

$$\chi \leq (1+k) \frac{\kappa q^m}{\alpha},$$

because the derivative races the two forces above. Extra space makes the child's space floor κ easier to meet, pushing fertility up; paying q^m for the additional space tightens the budget around the child's goods floor χ , pushing fertility down. The savings margin weakens the second force: the household can finance part of the larger housing bill by saving less, so only the fraction $1/(1+k)$ falls on current adult consumption, and the sign threshold scales up by $(1+k)$. A binding cap implies $\alpha x/s > q^m$, so the household-specific threshold is strictly above $(1+k)\kappa q^m/\alpha$. Hence the displayed condition is a uniform sufficient condition for positive fertility responses at all strictly binding caps. If the inequality fails, the response may still be positive for tightly constrained households, but $dn/dH < 0$ for binding caps sufficiently close to the unconstrained optimum.

This result maps into the model's access margins. A relaxation of the rental size cap raises $H = \bar{h}^R$. If the owner down-payment constraint is the unique binding component of the effective owner cap, then

$$\frac{\partial H}{\partial a_i} = \frac{\rho}{(1-\phi)q} > 0.$$

The derivative above signs the fertility response to these primitive cap relaxations locally, holding tenure, prices, income, and child costs fixed.

The competitive-equilibrium/planner comparison is a different statement because prices, consumption, tenure assignments, and entry can all change. In the competitive equilibrium, an entered young household in mode m satisfies, with $q^{CE,R} = q^{CE}$ and $q^{CE,O} = q^{CE} + \bar{\ell}/(1+r)$ under Assumption 1,

$$\frac{\vartheta(c_i^{CE,m} - \chi n_i^{CE,m})}{n_i^{CE,m}} = \chi + \kappa(q^{CE,m} + \zeta_i^{CE,m}).$$

A fertility-neutral planner has no direct social birth payoff. If the planner allocation for the same type and mode is interior with respect to any remaining access constraint, then

$$\frac{\vartheta(c_i^{P,m} - \chi n_i^{P,m})}{n_i^{P,m}} = \chi + \kappa Q^P.$$

Hence

$$n_i^{P,m} > n_i^{CE,m} \iff \frac{c_i^{P,m} - \chi n_i^{P,m}}{c_i^{CE,m} - \chi n_i^{CE,m}} > \frac{\chi + \kappa Q^P}{\chi + \kappa(q^{CE,m} + \zeta_i^{CE,m})}.$$

Thus a fertility-neutral planner does not mechanically deliver higher fertility merely because the competitive allocation has a positive access value. The mode-level result gives a primitive sufficient condition for a signed fertility response when a policy or planner locally raises a type's effective housing cap holding the case primitives fixed.

Under the baseline normalization $\bar{n}_i^E = 0$, births generated by the potential-young cohort are

$$N = \bar{M} \int \pi_i^E n_i^Y dG_Y(i).$$

If the outside option has fertility $\bar{n}_i^E$, then cohort births are instead

$$N = \bar{M} \int [\pi_i^E n_i^Y + (1 - \pi_i^E) \bar{n}_i^E] dG_Y(i),$$

and every entry-margin term below replaces n_i^Y by $n_i^Y - \bar{n}_i^E$. In the stationary demographic equilibrium, this birth mass and the renewal inflow satisfy $N + \bar{R} = \bar{M}$; the local comparisons below

hold $(\bar{M}, G_Y, F_O, F_R, \bar{R})$ fixed and do not by themselves trace the new stationary demographic fixed point after a policy change. Across two allocations,

$$N^{alt} - N^{CE} = \bar{M} \int \left[n_i^{alt}(\pi_i^{alt} - \pi_i^E) + \pi_i^E(n_i^{alt} - n_i^{CE}) \right] dG_Y(i),$$

where n_i^{CE} is fertility in the competitive-equilibrium inside mode and outside-option fertility is normalized to zero. The second term is the intensive fertility margin among the households who would enter in the competitive equilibrium. The first term is the entry margin induced by the change in inside value. If entry is held fixed, the aggregate fertility effect is just

$$N^{alt} - N^{CE} = \bar{M} \int \pi_i^E(n_i^{alt} - n_i^{CE}) dG_Y(i).$$

With entry held fixed, aggregate fertility rises only where the planner raises conditional fertility for constrained young types. With entry endogenous, higher inside values can also add entrants.

9.1 A local policy decomposition

The decomposition is fertility-neutral: it does not ask a planner to choose a target fertility level. It asks how a policy that relaxes a housing constraint, perhaps because the constraint misallocates the fixed stock, changes fertility as a by-product. The relevant levers act on primitive cap components. A down-payment grant raises a_i , and financial deregulation raises ϕ ; both loosen the collateral bound $a_i\rho/((1-\phi)q)$. Making family-sized units available to renters raises $\bar{h}^R$. Holding $\bar{H}$ fixed, raising $\bar{h}^R$ should be read as reclassifying, converting, or reallocating existing units into rental access, with any real conversion cost included in Ξ^Π ; the statistic changes access to the existing stock rather than aggregate supply.

Let z be a policy parameter. The local fixed-price effect depends on constrained exposure, pass-through to the binding cap, and the fertility response. For renters, $H_i^R = \bar{h}^R$. For owners,

$$H_i^O = \min \left\{ \bar{h}^O, \frac{a_i\rho}{(1-\phi)q} \right\}.$$

Let $m \in \{R, O\}$ denote the active tenure mode and let $\mathcal{E}_m(z)$ be the set of types for whom mode m is the unique active tenure, $H_i^m(z) < h_i^{u,m}$ is strictly binding, and the binding component of $H_i^m(z)$ is locally unique. Define the policy pass-through to the effective cap by

$$\mathcal{P}_i^m(z) = \frac{\partial H_i^m}{\partial z}.$$

Because the cap is a minimum, a policy that improves a slack component does nothing: a subsidy to units below the binding size does not relax a binding down-payment constraint, and a down-payment grant does nothing for a renter pinned by the rental size limit. If a policy does not relax the binding component of H_i^m , then $\mathcal{P}_i^m(z) = 0$ for this family-space-cap mechanism. Policies that operate primarily through the price q are a different exercise; they are outside this fixed-price formula, not evaluated by it. A property-tax capitalization exercise likewise isolates the down-payment channel only if $\bar{\ell}$ is held fixed; if $\bar{\ell} = \alpha_\ell \tau^q P \varpi / (1 + \varpi)$ moves with P or with the conversion factor α_ℓ , the policy also changes q^O and the old retention wedge. A transfer of spendable resources also moves fertility through the budget at a fixed cap; the decomposition isolates the access channel, and an instrument that both relaxes the cap and transfers resources adds the standard income term.

For each exposed type, write

$$x_i^m \equiv c_i^m - \chi n_i^m, \quad s_i^m \equiv h_i^m - \kappa n_i^m.$$

For capped types, the fixed-mode fertility response is

$$\Psi_i^m = \frac{\frac{\alpha\kappa}{(s_i^m)^2} - \frac{\chi q^m}{(1+k)(x_i^m)^2}}{\frac{\vartheta}{(n_i^m)^2} + \frac{\chi^2}{(1+k)(x_i^m)^2} + \frac{\alpha\kappa^2}{(s_i^m)^2}}.$$

The sharp pointwise sign condition is

$$\Psi_i^m > 0 \iff \chi < \frac{(1+k)\alpha\kappa(x_i^m)^2}{q^m(s_i^m)^2}.$$

A convenient uniform sufficient condition is $\chi \leq (1+k)\kappa q^m/\alpha$. The boundary case $\chi = (1+k)\kappa q^m/\alpha$ is equal scaling at lifetime prices: the child's consumption requirement matches its space requirement when adult consumption is priced at its lifetime cost $1+k$ and space at the mode price.

Let

$$\Theta_i^m \equiv \frac{\partial W_i^m}{\partial H_i^m} = \frac{\alpha}{s_i^m} - \frac{q^m}{x_i^m} = \frac{1}{x_i^m} \left[\frac{\alpha x_i^m}{s_i^m} - q^m \right]$$

be the utility value of relaxing the effective cap. The bracketed term is the bundle-implied access gap ζ_i^m , so this is the same object as ζ_i^m/x_i^m written without leaving an abstract multiplier in the formula.

Proposition 3 (Local fixed-state fertility sufficient statistic for family-housing access). *Holding prices, tenure choices, the active sets $\mathcal{A}$ —the collection of mode assignments and binding cap components—outside fertility, and the stationary demographic state fixed, the local aggregate fertility effect of a cap-shifting policy is*

$$\frac{dN}{dz} \Big|_{q, \text{tenure}, \mathcal{A}} = \bar{M} \sum_{m \in \{R, O\}} \int_{\mathcal{E}_m(z)} \left[\pi_i^E \Psi_i^m + (n_i^m - \bar{n}_i^E) \frac{\pi_i^E(1 - \pi_i^E)}{\kappa_E} \Theta_i^m \right] \mathcal{P}_i^m(z) dG_Y(i),$$

where $\bar{n}_i^E = 0$ under the baseline outside-option normalization. The first term is the intensive margin: already-entered constrained households adjust fertility by Ψ_i^m per unit of cap relief, positive under $\chi \leq (1+k)\kappa q^m/\alpha$. The second is the entry margin: relaxing a binding cap raises the value of entering the market, and entrants bring their fertility with them—net of the births they would have had outside. Both margins are gated by the same exposure set $\mathcal{E}_m(z)$: a policy aimed at unconstrained households moves nothing.

Corollary 1 (Property-tax capitalization, fixed user cost and state). *Consider a channel-isolating property-tax experiment that holds q , $\bar{\ell}$, tenure assignments, active sets, lifetime resources, outside fertility, and the current stationary state fixed. Let $\mathcal{E}_O^F$ be the set of types for whom ownership is the unique active tenure, the collateral component is the unique binding component of H_i^O , the owner-size limit is slack, and $H_i^O < h_i^{u,O}$. For these types,*

$$\frac{\partial H_i^O}{\partial \tau^p} \Big|_q = \frac{a_i}{(1-\phi)q}.$$

Therefore the capitalization channel changes births by

$$\frac{dN}{d\tau^p} \Big|_{q, \bar{\ell}, \text{tenure}, \mathcal{A}} = \bar{M} \int_{\mathcal{E}_O^F} \left[\pi_i^E \Psi_i^O + (n_i^O - \bar{n}_i^E) \frac{\pi_i^E(1 - \pi_i^E)}{\kappa_E} \left(\frac{\alpha}{s_i^O} - \frac{q^O}{x_i^O} \right) \right] \frac{a_i}{(1-\phi)q} dG_Y(i),$$

where

$$q^O = q + \frac{\bar{\ell}}{1+r}, \quad \Psi_i^O = \frac{\frac{\alpha\kappa}{(s_i^O)^2} - \frac{\chi q^O}{(1+k)(x_i^O)^2}}{\frac{\vartheta}{(n_i^O)^2} + \frac{\chi^2}{(1+k)(x_i^O)^2} + \frac{\alpha\kappa^2}{(s_i^O)^2}}.$$

The final factor is property-tax pass-through to the collateral cap. The bracketed term is the fertility and entry response per unit of cap relief. The market user cost q enters the down-payment pass-through through the asset price $P = q/\rho$, while the lifetime owner price q^O enters fertility and entry because it is the price relevant to the owner's bundle.

Remark 2 (Price feedback and endogenous gains exposure). *The corollary is a fixed-price impact derivative, not a full derivative between stationary equilibria. If the user cost adjusts, write reduced aggregate housing demand as $\tilde{H}^D(q, \tau^p) = \bar{H}$, after the direct household, entry, tenure, old-retention, and fiscal channels included in the experiment have been substituted in. Then*

$$\frac{dq}{d\tau^p} = -\frac{\tilde{H}_\tau^D}{\tilde{H}_q^D},$$

and the collateral-cap pass-through becomes

$$\frac{dH_i^F}{d\tau^p} = \frac{a_i}{(1-\phi)q} - \frac{a_i\rho}{(1-\phi)q^2} \frac{dq}{d\tau^p} = H_i^F \left[\frac{1}{\rho} - \frac{1}{q} \frac{dq}{d\tau^p} \right].$$

Thus the clean fixed- q statistic must be augmented by the equilibrium price pass-through. If the statutory gains levy moves with the asset price,

$$\ell_P = \tau^g \frac{\varpi}{1+\varpi} \frac{q}{\rho}, \quad \bar{\ell} = \mathbf{a}_\ell \ell_P,$$

then, when $\mathbf{a}_\ell$ is fixed,

$$\frac{d\bar{\ell}}{d\tau^p} = \bar{\ell} \left[\frac{1}{q} \frac{dq}{d\tau^p} - \frac{1}{\rho} \right].$$

With a moving conversion factor, add the term $\bar{\ell} d \log \mathbf{a}_\ell / d\tau^p$. In either case, the policy also changes $q^O = q + \bar{\ell}/(1+r)$ and the old release value $q - \bar{\ell}$. These are additional pass-through objects, not part of the clean cap-shifter sufficient statistic.

At the lifetime equal-scaling point, one observable object drives both margins.

Corollary 2 (Equal scaling and access gaps). *Suppose $\chi = (1+k)\kappa q^m/\alpha$, equal scaling at lifetime prices in mode m , and the effective cap binds strictly. Then*

$$\Psi_i^m = \frac{\kappa}{\alpha} \frac{\zeta_i^m}{c_i^m - \chi n_i^m} \Omega_i^m, \quad \Omega_i^m = \frac{\frac{\alpha}{h_i^m - \kappa n_i^m} + \frac{q^m}{c_i^m - \chi n_i^m}}{\frac{\vartheta}{(n_i^m)^2} + \frac{\chi^2}{(1+k)(c_i^m - \chi n_i^m)^2} + \frac{\alpha\kappa^2}{(h_i^m - \kappa n_i^m)^2}} > 0.$$

Under lifetime equal scaling, the access gap ζ_i^m measures both misallocation and fertility exposure. The entry margin runs on the same normalized access gap, since $\Theta_i^m = \zeta_i^m / (c_i^m - \chi n_i^m)$. Holding exposure, pass-through, entry probabilities, outside fertility, and the weight Ω_i^m fixed, larger normalized gaps predict larger birth responses.

Proof. At $\chi = (1+k)\kappa q^m/\alpha$, the numerator of Ψ_i^m is

$$\frac{\alpha\kappa}{(h_i^m - \kappa n_i^m)^2} - \frac{\kappa(q^m)^2}{\alpha(c_i^m - \chi n_i^m)^2} = \frac{\kappa}{\alpha} \left(\frac{\alpha}{h_i^m - \kappa n_i^m} - \frac{q^m}{c_i^m - \chi n_i^m} \right) \left(\frac{\alpha}{h_i^m - \kappa n_i^m} + \frac{q^m}{c_i^m - \chi n_i^m} \right) :$$

the $1+k$ in the child cost cancels the $1+k$ in the consumption term of the numerator. At a binding cap, $\alpha(c_i^m - \chi n_i^m)/(h_i^m - \kappa n_i^m) = q^m + \zeta_i^m$, so the first parenthesis equals $\zeta_i^m/(c_i^m - \chi n_i^m)$. Dividing by the positive denominator gives the expression above. $\square$

Thus the fixed-mode effect is exposure to a binding family-space constraint, times pass-through to the effective cap, times the fertility response, plus the induced entry response. The decomposition also holds tenure assignments fixed. With deterministic tenure choice, a policy that changes $W_i^O - W_i^R$ moves the indifference boundary $W_i^O = W_i^R$. If the type distribution has density near that boundary and fertility differs across cases, this tenure-switching margin is first order. The formula above is therefore the fixed-tenure component of the local fertility effect. The next subsection characterizes this margin. A full general-equilibrium policy experiment also changes the price, active sets, old housing choices, fiscal transfers, and possibly the resource cost Ξ^Π .

9.2 Tenure switching at the boundary

In this subsection q^R and q^O are tenure-mode effective user costs. The model now generates its own wedge: under Assumption 1, a buyer pays $q^O = q + \bar{\ell}/(1+r)$ per unit while a renter pays $q^R = q$; with no gains tax the modes price space identically. The anticipated tax also adds a constant to the owner mode's continuation—a locked-in old household occupies space at $q - \bar{\ell}$, an old-age subsidy worth $\beta\gamma \log(q/(q - \bar{\ell}))$ in value terms—which shifts tenure choice without moving any bundle. The restriction $\bar{\ell} < q$ in Assumption 1 keeps this constant well-defined. The proposition below carries all three case differences—effective price, cap, and continuation constant—explicitly; tenure-specific amenities, moving costs, or taste shifters still require a separate calculation.

Tenure switching is not a price or general-equilibrium effect. It is already present in the fixed-price model because tenure is chosen by

$$W_i^Y = \max\{W_i^R, W_i^O\}.$$

A policy that raises the owner value relative to the renter value moves households near the boundary $W_i^O = W_i^R$ from renting to owning, and because tenure is a discrete choice, the switchers' bundles—and possibly their fertility—jump. Whether they jump depends on a single feature of the environment: whether the two tenures price a unit of space differently. With $\bar{\ell} = 0$ they price it identically and the jump is exactly zero: the local policy formula of the previous subsection is then complete. With the tax, ownership is the expensive way to occupy space; the proposition gives conditions under which the boundary jump is signed. Further wedges—a rental premium for large units as in the smooth-cap formulation, landlord operating costs—move the gap the same way or against it.

The mode problems include savings. Concentrating the rule as above turns each mode into the static problem with consumption weight $1+k$ and mode price q^m , plus a mode constant: with common old preferences and interior old choices, the owner mode's continuation exceeds the renter's by

$$\Delta = \beta\gamma \log \frac{q}{q - \bar{\ell}} > 0$$

—the old-age subsidy of occupying retained space at $q - \bar{\ell}$ —and the constants coincide when $\bar{\ell} = 0$. The constant moves tenure choice without moving any bundle, so the proposition carries it explicitly: write K^A and K^B for the mode constants and $\Delta = K^B - K^A \geq 0$.

Fix prices and suppress the type index. Let A denote the cheaper mode and B the more expensive mode, with tenure-mode user costs

$$q^A < q^B.$$

Let (c_m, h_m, n_m) denote the mode- m optimum, $m \in \{A, B\}$. Let ζ^m be the goods-equivalent value of relaxing the effective family-housing cap in mode m .

Proposition 4 (Fertility at a tenure switch). *Suppose $q^A < q^B$, the mode continuations differ by $\Delta = K^B - K^A \geq 0$, and the household is indifferent, $W^A = W^B$, so the dear case runs a static-value deficit of exactly Δ . Then:*

(i) *if the price gap dominates the constant,*

$$\alpha \log \frac{q^B}{q^A} + \vartheta \log \frac{\chi + \kappa q^B}{\chi + \kappa q^A} > \Delta,$$

the cheaper mode's effective cap binds with positive value, $h_A = H^A$ and $\zeta^A > 0$;

(ii) *if moreover the wedge dominates the constant in resources,*

$$(q^B - q^A) h_B > \Delta (c_A - \chi n_A),$$

the switcher upsizes and trades consumption for space: $h_B > h_A$ and $c_B - \chi n_B < c_A - \chi n_A$;

(iii) *if in addition $\chi \leq (1 + k)\kappa q^A/\alpha$, then $n_B > n_A$ and $h_B - \kappa n_B > h_A - \kappa n_A$ whenever, with X, Y, Φ defined in the proof below,*

$$\int_{X_A}^{X_B} \frac{\Phi_X - \Phi_Y}{\Phi_Y} \Big|_{\Phi_A} dX > \int_{\Phi_A}^{\Phi_A + \Delta} \frac{d\varphi}{\Phi_Y(X_B, \cdot)}.$$

With $\Delta = 0$ the conditions in (i) and (ii) hold automatically, the level shift vanishes, and all conclusions reduce to the earlier ones.

Proof. Write $W^m = \tilde{W}^m + K^m$ up to common terms, where $\tilde{W}^m$ is the concentrated static value at price q^m and cap H^m ; indifference is $\tilde{W}^A - \tilde{W}^B = \Delta$.

(i) If $\zeta^A = 0$, mode A attains the unconstrained static value, and the closed forms give

$$\tilde{W}_{\text{unc}}(q^A) - \tilde{W}_{\text{unc}}(q^B) = \alpha \log \frac{q^B}{q^A} + \vartheta \log \frac{\chi + \kappa q^B}{\chi + \kappa q^A}.$$

Since $\tilde{W}^B \leq \tilde{W}_{\text{unc}}(q^B)$, the deficit would satisfy $\Delta \geq \tilde{W}_{\text{unc}}(q^A) - \tilde{W}_{\text{unc}}(q^B)$, contradicting price dominance. Thus $h_A = H^A$ and $\zeta^A > 0$.

(ii) Suppose $h_B \leq h_A$. The mode- B bundle is then feasible in mode A and costs $(q^B - q^A)h_B$ less. The static value is concave in resources with marginal utility $1/(c_A - \chi n_A)$ at w , so

$$\Delta = \tilde{W}^A - \tilde{W}^B \geq \frac{(q^B - q^A) h_B}{c_A - \chi n_A},$$

contradicting wedge dominance. Thus $h_B > h_A$.

Every mode optimum satisfies the same fertility condition as before. To see why the savings load does not multiply the child-goods term, write $x = c - \chi n$ and the concentrated resource constraint as $(1+k)x + \chi n + q^m h = w$. The FOC for x gives $(1+k)/x = \lambda(1+k)$, hence $\lambda = 1/x$; substituting this multiplier into the n condition gives

$$\frac{\vartheta}{n} = \frac{\chi}{c - \chi n} + \frac{\alpha \kappa}{h - \kappa n}.$$

Define

$$X = \frac{\chi}{c - \chi n}, \quad Y = \frac{\alpha \kappa}{h - \kappa n}.$$

X is large when adult consumption is squeezed, Y when adult space is squeezed.⁸ Then $n = \vartheta/(X + Y)$, and mode utility can be written as $K^m - \Phi(X, Y)$ up to constants common across modes because $\chi, \alpha, \kappa, \vartheta$ are fixed across tenure cases, where

$$\Phi(X, Y) = (1+k) \log X + \alpha \log Y + \vartheta \log(X + Y).$$

Indifference places the two optima on level sets that differ by the constant:

$$\Phi(X_B, Y_B) = \Phi(X_A, Y_A) + \Delta.$$

Occupied housing pins down the position along a level set: at any mode optimum,

$$h = \kappa \left(\frac{\alpha}{Y} + \frac{\vartheta}{X + Y} \right) = \kappa \Phi_Y(X, Y).$$

Let $T = X + Y$. Along a level set of Φ ,

$$\left. \frac{d\Phi_Y}{dX} \right|_{\Phi} = \frac{\Phi_{XY}\Phi_Y - \Phi_{YY}\Phi_X}{\Phi_Y},$$

and

$$\Phi_{XY}\Phi_Y - \Phi_{YY}\Phi_X = \frac{(1+k)\alpha}{XY^2} + \frac{(1+k)\vartheta}{XT^2} + \frac{\alpha\vartheta X}{Y^2T^2} > 0,$$

so on every level set Φ_Y is strictly increasing in X ; and at fixed X , moving to a higher level set raises Y and, since $\Phi_{YY} < 0$, lowers Φ_Y . Hence $h_B > h_A$ forces $X_B > X_A$, and $c_B - \chi n_B = \chi/X_B < \chi/X_A = c_A - \chi n_A$.

(iii) Write $T(X; \varphi)$ for $X + Y$ along the level set $\Phi = \varphi$, and decompose along the path that runs from (X_A, Y_A) along Φ_A to X_B , then up to $\Phi_A + \Delta$ at fixed X_B :

$$T_B - T_A = \underbrace{\int_{\Phi_A}^{\Phi_A + \Delta} \frac{d\varphi}{\Phi_Y(X_B, \cdot)}}_{\text{level shift } \geq 0} - \underbrace{\int_{X_A}^{X_B} \left. \frac{\Phi_X - \Phi_Y}{\Phi_Y} \right|_{\Phi_A} dX}_{\text{within-set force}}.$$

The within-set integrand is positive exactly where $(1+k)Y > \alpha X$, the region above the ray $Y = \alpha X/(1+k)$, which crosses each level set once: the level set is a decreasing graph, while along the ray $\Phi = (1+k + \alpha + \vartheta) \log X + \text{constant}$ is strictly increasing in X . At a mode optimum,

$$\frac{\alpha(c_m - \chi n_m)}{h_m - \kappa n_m} = q^m + \zeta^m, \quad \zeta^m \geq 0, \quad \text{so} \quad \frac{Y_m}{X_m} = \frac{\kappa(q^m + \zeta^m)}{\chi},$$

⁸Capital X and Y are local to this proof and unrelated to income y_i .

and under $\chi \leq (1+k)\kappa q^A/\alpha$ both optima lie strictly in the region: mode A because $\zeta^A > 0$, mode B because $q^B > q^A$. Then $n_B = \vartheta/T_B > \vartheta/T_A = n_A$ if and only if the within-set force exceeds the level shift. When it does, $T_B < T_A$ together with $X_B > X_A$ gives $Y_B < Y_A$, hence $h_B - \kappa n_B = \alpha\kappa/Y_B > h_A - \kappa n_A$. With $\Delta = 0$ the level shift vanishes, the path stays in the region between the two optima, and the inequality always holds. $\square$

At indifference there are now two reasons to pay more per unit of space: to get more space, and to buy the dear mode's better old age. The conditions price that second temptation: the wedge must dominate the subsidy in resources for the switcher to upsize, and the fertility jump nets the within-mode space force against the subsidy's level shift. For the separate boundary calculation, not the worked local allocation above, the conditions hold with room to spare—price dominance by a factor of four—and the boundary household upsizes from 0.15 to 0.20 and adds 0.007 children.⁹

The rent-own cases now read off directly, with the effective user costs $q^R = q$ and $q^O = q + \bar{\ell}/(1+r)$. *No gains tax* ($\bar{\ell} = 0$, so $q^O = q^R = q$): the modes differ only through their effective caps, and an indifferent household has the same allocation in both—either both caps are slack, or the caps coincide at the binding allocation. Then $n_O = n_R$, there is no discrete jump at the boundary, and the fixed-tenure formula of the previous subsection is the complete fixed-price answer. *With the tax* ($q^O > q^R$): ownership is the expensive family-space mode; when the dominance conditions and part (iii)'s inequality hold, an indifferent switcher into ownership upsizes and has $n_O > n_R$. *Renting dearer* ($q^O < q^R$, for example a large-unit rental premium): the sign is not implied by the proposition unless the dearer rental mode also has weakly higher mode constant, or a separate lemma allows $\Delta < 0$. A cost wedge between the tenures is what makes tenure switching a first-order fertility-jump margin, but $q^O > q^R$ alone does not sign the jump without the proposition's conditions.

The switching formula is fixed-price accounting, not a welfare result or a general-equilibrium comparative static. Let types live in a smooth finite-dimensional type space with continuous density $f(i)$, and suppose there are no mass points at the tenure boundary. Let

$$\Delta_i(z) = W_i^O(z) - W_i^R(z), \quad \mathcal{I}(z) = \{i : \Delta_i(z) = 0\},$$

and assume $\nabla_i \Delta_i(z) \neq 0$ on $\mathcal{I}(z)$, with unique mode optima and kinks in active constraints having zero measure. The switching component of the fixed-price fertility effect is

$$\left. \frac{dN^{\text{switch}}}{dz} \right|_{q,\mathcal{A}} = \bar{M} \int_{\mathcal{I}(z)} \pi_i^E (n_i^O - n_i^R) \frac{\partial_z \Delta_i(z)}{\|\nabla_i \Delta_i(z)\|} f(i) dS(i),$$

where $dS(i)$ is surface measure on the indifference boundary. The normal is oriented toward the owner region $\{\Delta_i > 0\}$, so positive $\partial_z \Delta_i$ moves renter-boundary mass into ownership. In the boundary terms below, $x_i^m = c_i^m - \chi n_i^m$ and $s_i^m = h_i^m - \kappa n_i^m$ are evaluated at the mode- m optimum for the boundary type. For cap-shifting policies,

$$\partial_z \Delta_i(z) = \left(\frac{\alpha}{s_i^O} - \frac{q^O}{x_i^O} \right) \mathcal{P}_i^O(z) - \left(\frac{\alpha}{s_i^R} - \frac{q^R}{x_i^R} \right) \mathcal{P}_i^R(z).$$

The total fixed-price effect is the sum of the fixed-tenure sufficient statistic and the boundary term:

$$\left. \frac{dN}{dz} \right|_q = \left. \frac{dN}{dz} \right|_{q,\text{tenure},\mathcal{A}} + \left. \frac{dN^{\text{switch}}}{dz} \right|_{q,\mathcal{A}}.$$

⁹Boundary calculation: use the worked-example preference and tax parameters, so $\alpha = 0.5$, $\vartheta = 0.4$, $\kappa = 0.3$, $\chi = 0.6$, $k = 0.26445$, $q^A = q^R = 1$, $q^B = q^O = 1.14815$, and $\Delta = \beta\gamma \log(q/(q-\bar{\ell})) = 0.01967$. Set caps $H^A = 0.15$ and $H^B = 0.20$, and choose lifetime resources $w = 0.75304$ so the type is indifferent. The capped optima are $n_A = 0.14025$ and $n_B = 0.14769$.

No separate entry-boundary term appears under logit entry because the inside value is continuous at the rent-own boundary; the discontinuity in births is the conditional fertility jump $\pi_i^E(n_i^O - n_i^R)$.

Corollary 3 (Owner-access policies). *Consider an owner-access policy with $\mathcal{P}_i^O(z) \geq 0$ and $\mathcal{P}_i^R(z) = 0$. Suppose the conditions of Proposition 4, including part (iii)'s inequality, hold for almost every type on $\mathcal{I}(z)$. If $q^O > q^R$, the switching component is nonnegative, so the fixed-tenure owner-access formula is a lower bound for this fixed-price fertility channel. If $q^O = q^R$ (the no-tax/no-tenure-price-wedge benchmark), the switching component is zero. If $q^O < q^R$, the sign is not pinned by this proposition unless the alternative environment satisfies its dear-mode-constant condition.*

Corollary 4 (Rental-size limit policies). *Consider a rental-size limit policy with $\mathcal{P}_i^R(z) \geq 0$ and $\mathcal{P}_i^O(z) = 0$. Suppose the conditions of Proposition 4, including part (iii)'s inequality, hold for almost every type on $\mathcal{I}(z)$. If $q^O > q^R$, the switching component is nonpositive: relaxing the rental cap raises fertility for constrained renters who remain renters, but it can also pull marginal owners back into renting. If $q^O = q^R$ (the no-tax/no-tenure-price-wedge benchmark), the switching component is zero. If $q^O < q^R$, the sign is not pinned by this proposition unless the alternative environment satisfies its dear-mode-constant condition.*

References

- Barro, Robert J., and Gary S. Becker. 1988. “A Reformulation of the Economic Theory of Fertility.” *Quarterly Journal of Economics* 103(1): 1–25.
- Barro, Robert J., and Gary S. Becker. 1989. “Fertility Choice in a Model of Economic Growth.” *Econometrica* 57(2): 481–501.
- Coven, Joshua, Sebastian Golder, Arpit Gupta, and Abdoulaye Ndiaye. 2025. “Property Taxes and Housing Allocation Under Financial Constraints.” Working paper, June 2025. Earlier version: CESifo Working Paper No. 11203, 2024.
- Diamond, Peter A. 1965. “National Debt in a Neoclassical Growth Model.” *American Economic Review* 55(5): 1126–1150.
- Fonseca, Julia, Lu Liu, and Pierre Mabille. 2024. “Unlocking Mortgage Lock-In: Evidence from a Spatial Housing Ladder Model.” Working paper.
- Galor, Oded, and David N. Weil. 1996. “The Gender Gap, Fertility, and Growth.” *American Economic Review* 86(3): 374–387.
- Gervais, Martin. 2002. “Housing Taxation and Capital Accumulation.” *Journal of Monetary Economics* 49(7): 1461–1489.
- Sommer, Kamila, and Paul Sullivan. 2018. “Implications of US Tax Policy for House Prices, Rents, and Homeownership.” *American Economic Review* 108(2): 241–274.